

Md Abdus Sami Akanda

MS in Physics, University of Nebraska-Lincoln — Researcher — Academic Instructor
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AREAS OF RESEARCH INTEREST

Broader Area: Condensed Matter Physics

Specific Areas:

- Spintronics
- Magnetization/Domain Wall Dynamics
- Magnetic Nanowire
- Landau-Lifshitz-Gilbert Equation
- Spin-Transfer Torque
- Spin-Orbit Torque
- Spin Waves
- Magnonic Crystal

EDUCATION

University of Nebraska-Lincoln

Master of Science in Physics
CGPA - 3.39 out of 4.00

Nebraska, USA
August, 2023 - December, 2025

Khulna University

Bachelor of Science in Physics
CGPA - 3.34 out of 4.00

Khulna, Bangladesh
January, 2018 - February, 2023

Cantonment Public School and College

Higher School Certificate (Science)
CGPA - 5.00 out of 5.00

Mymensingh, Bangladesh
July, 2015 - July, 2017

WORK EXPERIENCES

University of Nebraska-Lincoln

Graduate Teaching Assistant

Lincoln, Nebraska, USA
August, 2023 – December, 2025

Khulna University

Research Assistant, Physics Discipline

Khulna, Bangladesh
September, 2020 - November, 2022

Udvash Academic & Admission Care

Mathematics Advisor

Khulna, Bangladesh
May, 2018 – July, 2019

PUBLICATIONS

“Alternating current triggered exchange-mediated picosecond switching of cylindrical magnetic tunnel junctions”, *M. A. Jasem, Y. D. Liu, M. T. Islam, M. A. S. Akanda, X. R. Wang, and T. Min. Journal of Applied Physics - Under Review* 2026

“Localized thermal gradient driven domain wall dynamics in ferromagnetic nanowire”, *M. A. J. Pikul, M. A. S. Akanda, and M. T. Islam. Physica B: Condensed Matter - Under Review* 2026

“Thermal effect on microwave pulse-driven magnetization switching of Stoner particle”, *S. Chowdhury, M. A. S. Akanda, M. A. J. Pikul, M. T. Islam, and T. Min. Physica Scripta, Institute of Physics (Q2; IF: 2.9) (DOI: 10.1088/1402-4896/ad1706).* 2023

“Role of SSW on thermal-gradient induced domain-wall dynamics”, *M. A. S. Akanda, M. T. Islam, and X. R. Wang. Journal of Physics: Condensed Matter, 35(31), 315701, Institute of Physics (Q1; IF: 2.745) (DOI: 10.1088/1361-648X/acfcde).* 2023

“Role of shape anisotropy on thermal gradient-driven domain wall dynamics in magnetic nanowires”, *M. T. Islam, M. A. S. Akanda, F. Yesmin, M. A. J. Pikul, and J. M. T. Islam. Modern Physics Letters B, 37(12), 2350013, World Scientific Publishing (Q3; IF: 1.9) (DOI: 10.1142/S0217984923500136).* 2023

“Shape anisotropy effect on magnetization reversal induced by linear down chirp pulse”, *Z. K. Juthy, M. A. J. Pikul, M. A. S. Akanda, M. T. Islam. Physica B: Condensed Matter, 630, 413671, Elsevier (Q2; IF: 2.988) (DOI: 10.1016/j.physb.2022.413671).* 2022

“Fast magnetization reversal of a magnetic nanoparticle induced by cosine chirp microwave field pulse”, *M. T. Islam, M. A. S. Akanda, M. A. J. Pikul and X. S. Wang. Journal of Physics: Condensed Matter, 34(10), 105802, Institute of Physics (Q1; IF: 2.745) (DOI: 10.1088/1361-648X/ac3f66).* 2021

CONFERENCE PROCEEDINGS

“Shape anisotropy effect on magnetic domain wall dynamics in nanowires under thermal gradient”, *M. T. Islam, M. A. S. Akanda, F. Yesmin, M. A. J. Pikul, and J. M. T. Islam*. International Conference on STEM and the Fourth Industrial Revolution 2022. July 1-3, 2022

“Fast magnetization reversal of a magnetic nanoparticle driven by a chirp microwave field pulse at finite temperature”, *M. T. Islam, M. A. J. Pikul, M. A. S. Akanda, and X. S. Wang*. National Conference on Physics - 2021 P13. August 6-7, 2021

“A Strategy of Swift Magnetization Reversal of Single Domain Magnetic Nanoparticle”, *M. A. S. Akanda, M. A. J. Pikul, Z. K. Juthy, M. T. Islam, and X. S. Wang*. International e-Conference on Physics 2021 P05. July 9-11, 2021

“Temperature Effect on Magnetization Reversal of a Magnetic Nanoparticle Driven by a Down-Chirp Microwave Field Pulse”, *M. A. J. Pikul, M. A. S. Akanda, Z. K. Juthy, M. T. Islam, and X. S. Wang*. International e-Conference on Physics 2021 P05. July 9-11, 2021

“Influence of shape anisotropy on magnetization reversal induced by non-linear down chirp pulse”, *Md. Abdus Sami Akanda*. 2021 ESpinRed School on Spintronics. May 26, 2021

“LLG equation, non-linear dynamics of magnetization reversal and shape anisotropy”, *Md. Abdus Sami Akanda*. Physics Talk 2021. March 14, 2021

POSTER PRESENTATIONS

“Spin Wave Dispersion Relation Calculations in a Permalloy Microstrip” *S. K. Saha, M. A. J. Pikul, and M. A. S. Akanda*. International Conference on Physics 2026. April 10, 2026

“Fast magnetization reversal of a magnetic nanoparticle driven by a down-chirp microwave field pulse at finite temperature” *M. A. J. Pikul, M. A. S. Akanda, M. T. Islam, and X. S. Wang*. VI International Workshop on Dzyaloshinskii-Moriya Interaction and Exotic Spin Structures (DMI-2021). September 6-10, 2021

SOFTWARE SKILLS

Operating Systems: Mac, Windows, Linux
Micromagnetic Simulation: MuMax3, COMSOL
Programming: C, LaTeX, Python, MATLAB, JavaScript
Data analysis & Plotting: OriginLab
Microsoft Office: MS Word, MS Excel, MS PowerPoint
Graphic: Adobe Illustrator, Blender
Web Development: HTML, CSS, React.js

See the projects on Github.

ADDITIONAL ACADEMIC COURSES

Quantum Mechanics & Quantum Computing: COMSOL & Qisqit June 3, 2022 - September 9, 2022
Instructed by Dr. Mahdy Rahman Chowdhury, Associate Professor, North South University, Dhaka, Bangladesh

2021 ESpinRed School on Spintronics May 17, 2021 - May 31, 2021
Organised by Dr. Fernando Bartolomé, INMA, CSIC - University of Zaragoza, Spanish Network on Spintronics, Spain

WORKSHOPS & SEMINARS

Online Spintronics Seminar Series March 27, 2020 - December 12, 2025
Organised by Prof. Kirill Belashchenko (UNL), USA and Prof. Xin Fan (DU), USA

Professional and Communications Training for Scientists May 23, 2021 - May 27, 2022
Organized by Kelsey Calhoun, ICTP, Trieste, Italy

Young Researchers’ Workshop on Machine Learning for Materials May 9, 2021 - May 13, 2022
Organized by Prof. Sandro Scandolo, ICTP, Trieste, Italy

2D Materials for Spin-Orbitronics May 3, 2021 - May 5, 2021
Organized by Prof. Nadia Binggeli, ICTP, Trieste, Italy

2nd Workshop on Classical Electromagnetism: Tesla’s Gift October 18, 2020 - October 18, 2020
Organized by Community of Physics, Dhaka, Bangladesh

Mumax3 Workshop	August 31, 2020 - September 14, 2020
Instructed by Dr. Jonathan Leliaert and Dr. Jeroen Mulkers, DyNaMat group, Ghent University, Belgium	
Seminar on “Research and How to Build Up an Academic Career by it”	October 17, 2019
Organized by Khulna University, Khulna, Bangladesh	
1st Workshop on Differential Geometry: Einstein’s Insight	June 15, 2019 - June 21, 2019
Organized by Community of Physics, Dhaka, Bangladesh	
Conference on “Weather Forecasting and Advances in Physics”	May 11, 2018
Organised by Khulna University of Engineering & Technology, Khulna, Bangladesh	

AWARDS AND ACHIEVEMENTS

International Conference on STEM and the Fourth Industrial Revolution, 2022	July 3, 2022
Best Paper Award	
The 2021 University Physics Competition	January, 2022
Silver Medal Rank	
International Theoretical Physics Olympiad	August, 2021
16th Position	
The 2020 University Physics Competition	January, 2021
Silver Medal Rank	
The 23rd International Rudolf Ortway Problem Solving Contest in Physics	December, 2020
11th Position, 3rd Year Category	
International Theoretical Physics Olympiad	April, 2020
21th Position	
The 2019 University Physics Competition	January, 2020
Silver Medal Rank	

CO-CURRICULAR ACTIVITIES

Institute for Theoretical and Computational Physics Research	January, 2024 - Present
Director of Operations and Co-Founder	
Fermion Physics Club	May, 2019 - January, 2025
Instructor and Founder	
Bangladesh Youth Environmental Initiative (BYEI)	July, 2019 - February, 2023
Regional Coordinator, Khulna, Bangladesh	
Bangladesh Olympiad on Astronomy and Astrophysics (BDOAA)	March, 2018 - December, 2022
Divisional Coordinator, Khulna, Bangladesh	

LANGUAGES

- Bengali (Native)
- English (Proficient)

TEST SCORES

IELTS (Academic): Test Date: November 10, 2022
Overall-7.5, Listening-8.5, Reading-8.0, Writing-7.0, Speaking-7.0

REFEREES

Dr. Md. Torikul Islam Professor Physics Discipline Khulna University, Bangladesh Email: torikul@phy.ku.ac.bd Personal Webpage	Dr. Md. Omar Faruk Rasel Associate Professor Physics Discipline Khulna University, Bangladesh Email: ofr_ju@phy.ku.ac.bd Personal Webpage
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